

Components for Optical Communication Systems (cont.)

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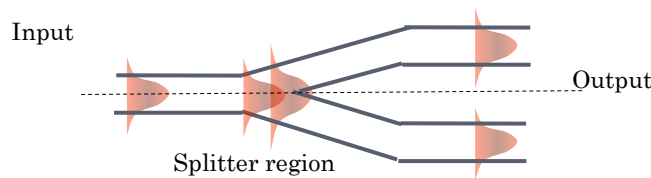
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The Y junction

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The Y junction

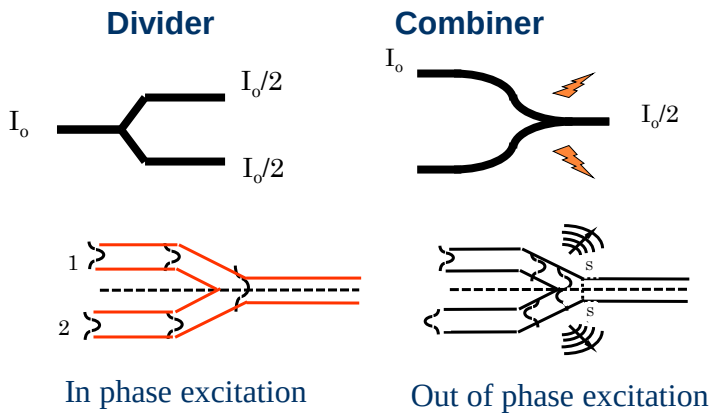


The Y junction is the simplest compact splitter that satisfy:

- Wide bandwidth (broad operating wavelength range)
- Low insertion loss
- Low non-uniformity in the case of symmetric Y junction
- High reliability
- Minimal dimensions (compact)

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The Y junction combiner



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The Y junction combiner

A symmetric Y junction can also be used as an optical combiner.

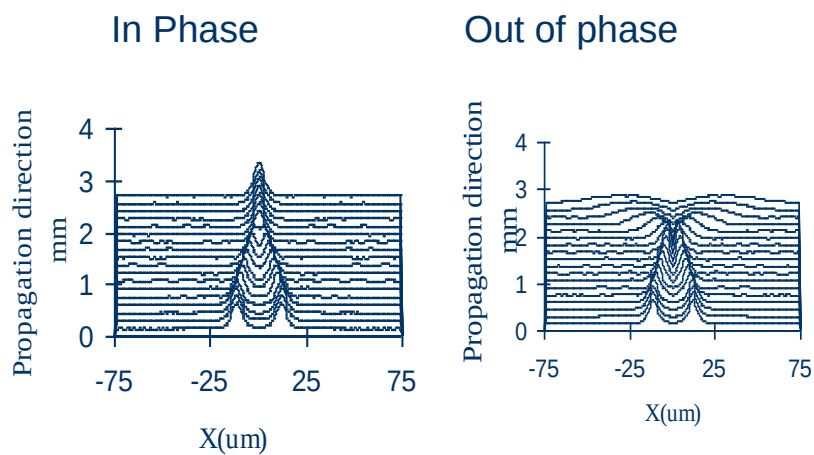
*In this case, when the two input of the junction are excited by two in phase optical signals, the output is the intensity sum of the two inputs.

*However, if the two inputs are excited by two out of phase signals, an odd mode is excited in the transition region and when the input/output waveguides of the junction are all single mode guides, the output is radiated in the substrate.

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The Y junction combiner



Y JUNCTION EXCITATION

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The Y junction combiner

Previous figure shows the simulation of the in-phase and out of phase excitation of an integrated optical Y junction.

We can see how the in-phase excitation leads to the construction of the even fundamental mode of the input guide while the out of phase excitation leads to the formation of the odd mode in the transition region and as the output guide is SM the power will radiate in the substrate.



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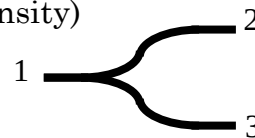
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TRANSMISSION COEFFICIENT

- As a power divider, from symmetry:

$$\rightarrow I_2 = I_3 = I_1/2 \quad (I: \text{Intensity})$$

$$\rightarrow E_2 \text{ \& } E_3 \text{ are in phase}$$



- As a power combiner

$$T_{21} = T_{31} = \frac{1}{\sqrt{2}} \quad T: \text{Electric field transmission coefficient}$$

– If $E_2 = E_3$ and in phase

$$\rightarrow E_1 = \frac{E_2 + E_3}{\sqrt{2}}$$

Such that: $I_1 = I_2 + I_3$



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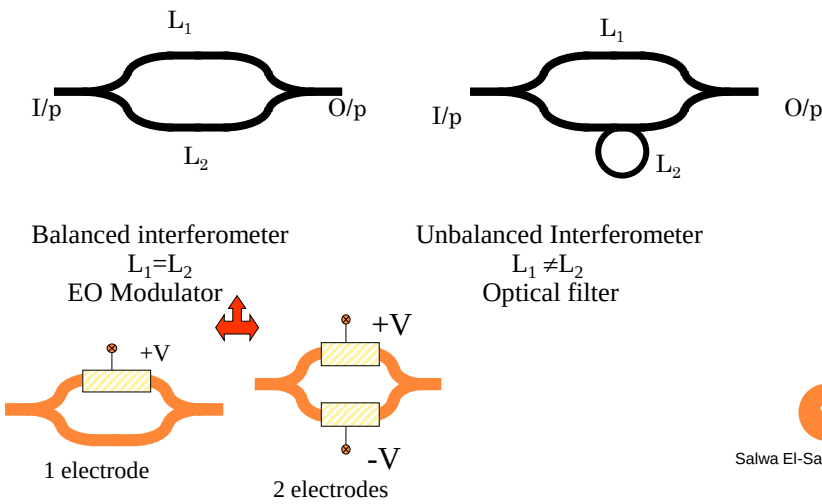
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Mach-Zehnder Interferometer

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Mach-Zehnder Interferometer



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Mach-Zehnder Interferometer

*A Mach-Zehnder interferometer could be a balanced or unbalanced one. The interferometer is called balanced when the two arms of the interferometer have equal lengths.

***An Unbalanced interferometer is characterized by a path difference between the two arms $dL = L_1 - L_2$. This path difference results in a wavelength dependent phase shift between the two arms and thus the unbalanced MZ interferometer could be used as an optical filter.**

*When the MZ is balanced, such phase shift could be produced by an external applied voltage (in case of electrooptic modulator, for example), and thus the output signals varies according to the applied electrical signal.

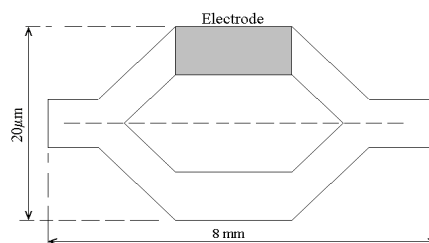
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Electrooptic modulator

External modulator

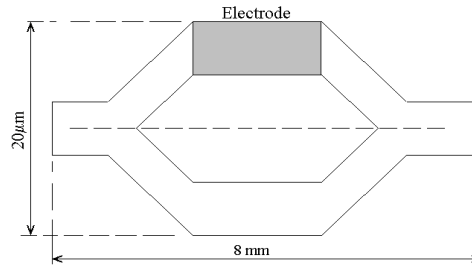
- Electro-Absorption EA modulator $\rightarrow \alpha = \alpha(V_{ap})$
 - Control the material absorption by the applied voltage
- Electro-optics EO Modulator $\rightarrow n = n(V_{ap})$
 - Control the optical phase shift by the applied voltage



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Electrooptic modulator



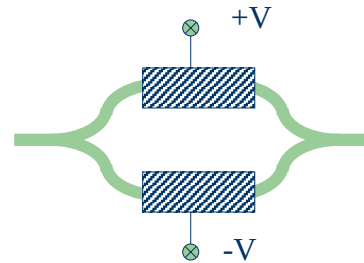
The Idea of an Electrooptic modulator is based on the following issues:

- 1- Light is split into two isolated (non-interacting) waveguides.
- 2- Applied electric field from electrode modifies material refractive index and so it modifies relative phase between light in the two arms.
- 3- The interferometer transforms the phase modulation into intensity modulation.

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EO Modulator



*The principle of operation is based on the electro-optic effect: a change of optical refractive index in the waveguide due to application of an electric field. A light beam traveling in the input optical waveguide is split into two beams, propagating a suitable distance and then recombined, giving rise to interference.

*The interference is constructive when the two beams are in phase, and the full power exits to the output waveguide. If the optical lengths of the two arms differ **by half of a wavelength (optical phase of π)**, the interference is destructive. A change of the refractive index of the waveguide resulting from the application of an electrical signal to the electrode causes an optical phase shift, hence intensities change in the outputs of the interferometer

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EO Modulator

Electro Optic Mach-Zehnder Modulator

Electro-optics EO effect

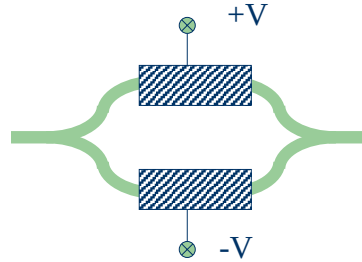
$$\rightarrow n = n(V_{ap})$$

$$\rightarrow n \Rightarrow n \pm \Delta n$$

$$\beta \Rightarrow \beta \pm \Delta\beta$$

$$E_{op} = \frac{E_{in}}{2} [e^{-j\beta_1 L} + e^{-j\beta_2 L}]$$

$$E_{op} = \frac{E_{in}}{2} [e^{-j(\beta - \Delta\beta)L} + e^{-j(\beta + \Delta\beta)L}] = E_{in} e^{-j\beta L} \cos(\Delta\beta L)$$



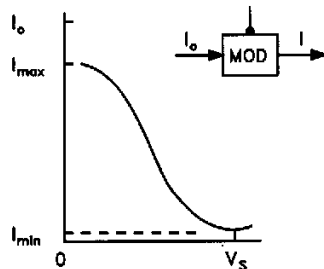
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Modulator Response

Switching Response



$$\text{Insertion loss (dB)} = 10 \log_{10} (I_{\max}/I_0)$$

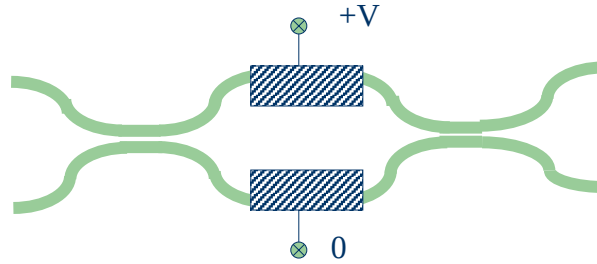
$$\text{Extinction ratio (dB)} = -10 \log_{10} (I_{\min}/I_{\max})$$

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MZ Modulator as a switch



To use the MZ modulator as a switch, the y junction at the input and output could be replaced by a directional coupler. The voltage will control which output of the coupler will have 1 and which output will have 0

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Unbalanced MZ Interferometer

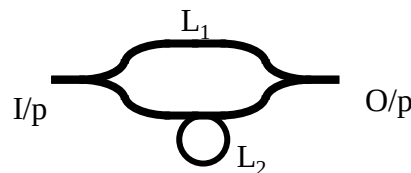
Unbalanced Interferometer

$$L_1 \neq L_2$$

Optical filter

$$L_{av} = (L_1 + L_2)/2$$

$$\Delta L = L_1 - L_2$$



$$E_{op} = \frac{E_{in}}{2} e^{-j\beta L_1} \left[1 + e^{j\beta(L_1 - L_2)} \right] = E_{in} e^{-j\beta L_{av}} \cos\left(\frac{\beta \Delta L}{2}\right)$$

$$I_{op} = I_{in} \cos^2\left(\pi \frac{n_{eff} \Delta L}{\lambda}\right)$$

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UBMZI Filter Response

Periodic filter response

$$I_{op} = I_{in} \cos^2 \left(\pi \frac{n_{eff} \Delta L}{\lambda} \right)$$

- FSR: $\Delta \nu = C / \Delta L n_{eff}$

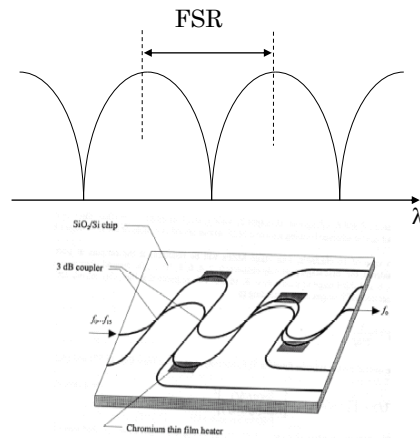
$$\Delta \lambda = \lambda^2 / n_{eff} \Delta L$$

- For max. $I_{out} = I_{in}$

- For min. $I_{out} = 0$

Cascaded MZ interferometers

- Heaters could be used for tuning



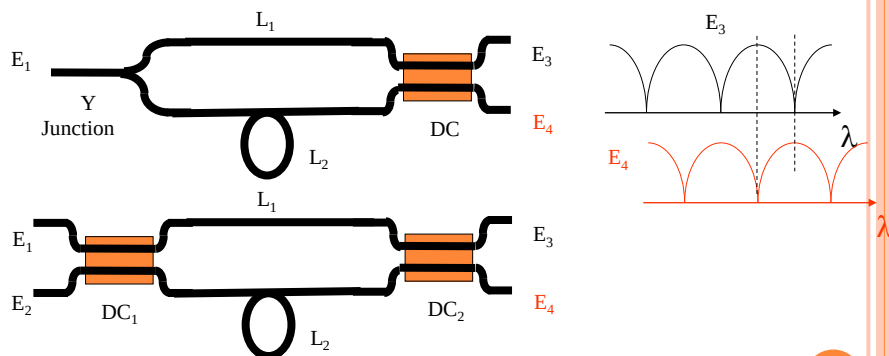
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Two Outputs MZ Interferometer

- When a directional coupler is used at the output of the MZ interferometer we get two complementary outputs.



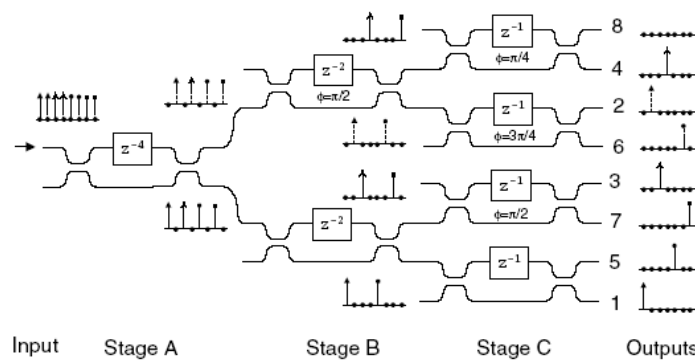
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Application: WDM Mux

- Use a repeated unit cell with different coefficients
- Ex → 1x8 DeMux



$$Z = e^{j\beta\Delta L}$$

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1x8 optical demultiplexer

Using successive MZ interferometers to build a 1x8 optical demultiplexer.

*Assuming the input wavelengths to be corresponding to channels 1, 2, 3, 4, 5, 6, 7, 8. The first MZ interferometer from left can be used to separate the wavelengths corresponding to channels 1, 3, 5, 7 in one output and the wavelengths corresponding to channels 2, 4, 6, 8 in another output using the periodicity of the MZI in stage A.

*Then in stage B the channels 1, 3, 5, 7 can again be multiplexed to 1, 5 in one output and 3, 7 in another output using again the new periodicity of the MZ in this stage. Using the same process, stage C will be finally able to deliver each channel in a separate waveguide.

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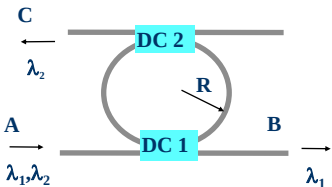
Ring Resonator



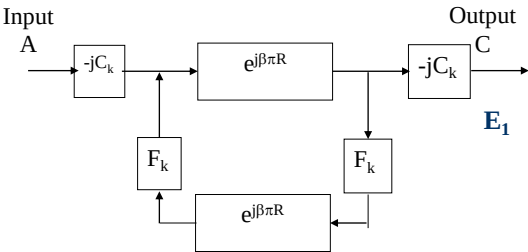
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Ring Resonator

$$T_{eq} = -C_k^2 \frac{e^{j\pi\beta R}}{1 - F_k^2 e^{j2\pi\beta R}}$$



$$F_k^2 = 1 - C_k^2$$
$$C_k = C_k(\text{gap})$$



$$E_1 \rightarrow \text{DC 1} \rightarrow E_3 = aE_1 = F_k E_1$$
$$E_4 = -jbE_1 = -jC_k E_1$$



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Ring Resonator

*This actually represents the guided wave equivalent form of the well known Fabry-Perot interferometer where, as shown in the previous Fig., the input optical signal coupled to the resonator, is forced to make infinite number of round trips inside the resonator with a fraction of it coupled to the external guides at each round trip.

*The main performance characteristics of these resonators can thus be deduced in analogy with the FP resonator.

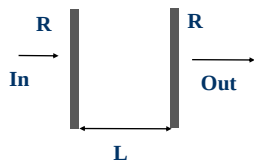
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Analogy with Fabry Perot Interferometer

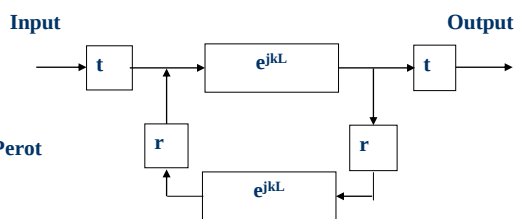
• Response



Structure of a symmetric Fabry-Perot

$$\begin{aligned} 2L &\Rightarrow 2\pi R \\ r &\Rightarrow F_k \\ t &\Rightarrow -j C_k \end{aligned}$$

$$t_{eq} = t^2 \frac{e^{jkL}}{1 - r^2 e^{j2kL}}$$



Block diagram representation

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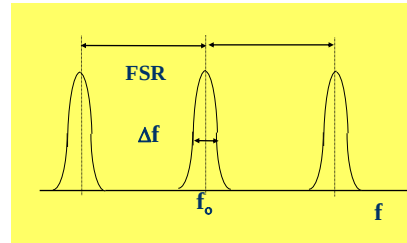
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Lossless Ring filter

- Filter parameters

- Response:

$$\rightarrow I_{op} = I_{in} \frac{1}{1 + \left(\frac{4F_k^2}{(1 - F_k^2)^2} \right) \sin^2(\beta\pi R)}$$



- Max. =1 @ $k\pi R = q\pi \rightarrow FSR = \frac{C}{2n_{eff}\pi R}$

$$FSR_f = \frac{C}{2n_{eff}\pi R} \Leftrightarrow FSR_\lambda = \frac{\lambda^2}{2n_{eff}\pi R}$$

- FWHM @ $k\pi R = q\pi + \theta \rightarrow T_{eq}(q\pi + \theta) = 1/2 \rightarrow \Delta f = \frac{C}{2n_{eff}\pi R} \frac{1 - F_k^2}{\pi F_k}$

- Finesse $F = FSR/\Delta f \rightarrow Finesse = \frac{\pi F_k}{1 - F_k^2}$

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Ring Response

- Typical Lossless response:

- Max. $\rightarrow T_{max} = 1$

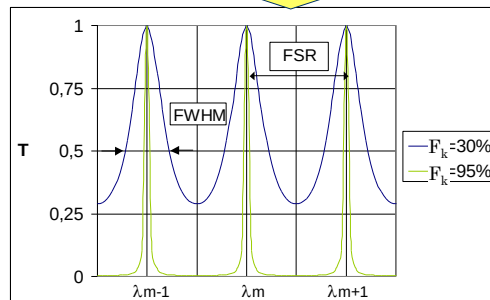
- Min. $\rightarrow T_{min} = \frac{(1 - F_k^2)^2}{(1 + F_k^2)^2}$

- $FSR = \frac{C}{2n\pi R}$

$$FSR = \Delta \nu = \frac{C}{2n_{eff} \pi R}$$

$$\Rightarrow \Delta \lambda = \frac{\lambda^2}{2n_{eff} \pi R}$$

$$T_{eq} = \frac{1}{1 + \left(\frac{4F_k^2}{(1 - F_k^2)^2} \right) \sin^2(\beta\pi R)}$$



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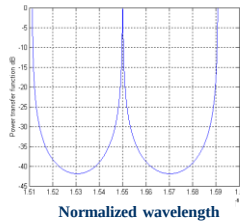
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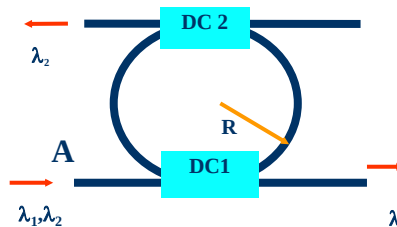
Ring Filter Response

• Notch Filter

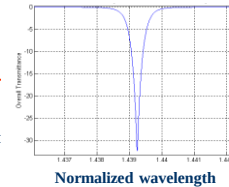
Typical Transmission Response



Selectivity $\rightarrow$ Coupling + Losses
FSR $\rightarrow$ Diameter



Typical Reflection Response



Design parameters

$\rightarrow$ -Coupling ratio : Gap between ring & straight guide
& -Ring radius

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Ring Filter Response

*If an optical signal is injected at port A, this signal may be in resonance in the ring and then couples out from the resonator into the upper fiber waveguide, and exits the device at port C. This corresponds to the transmitted signal in the FP resonator. A signal that is off-resonance with the cavity remains in lower fiber and exits at port B in a way similar to the reflected signal in the FP.

*The input and output coupling ratios between the straight guides and the ring are thus equivalent to the reflectivities of the Fabry–Perot resonator mirrors. The finesse of the resonator depends mainly on these coupling ratios as well as the propagation loss while the FSR is given by $FSR = C/2\pi R$ with R the radius of the ring.

*The critical dimension in this filter is the gap separating the microring cavity from the tangential waveguides. These gap sizes determine the input and output coupling ratios of the resonator, which in turn determine the magnitude of the finesse and the at-resonance transmission.

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